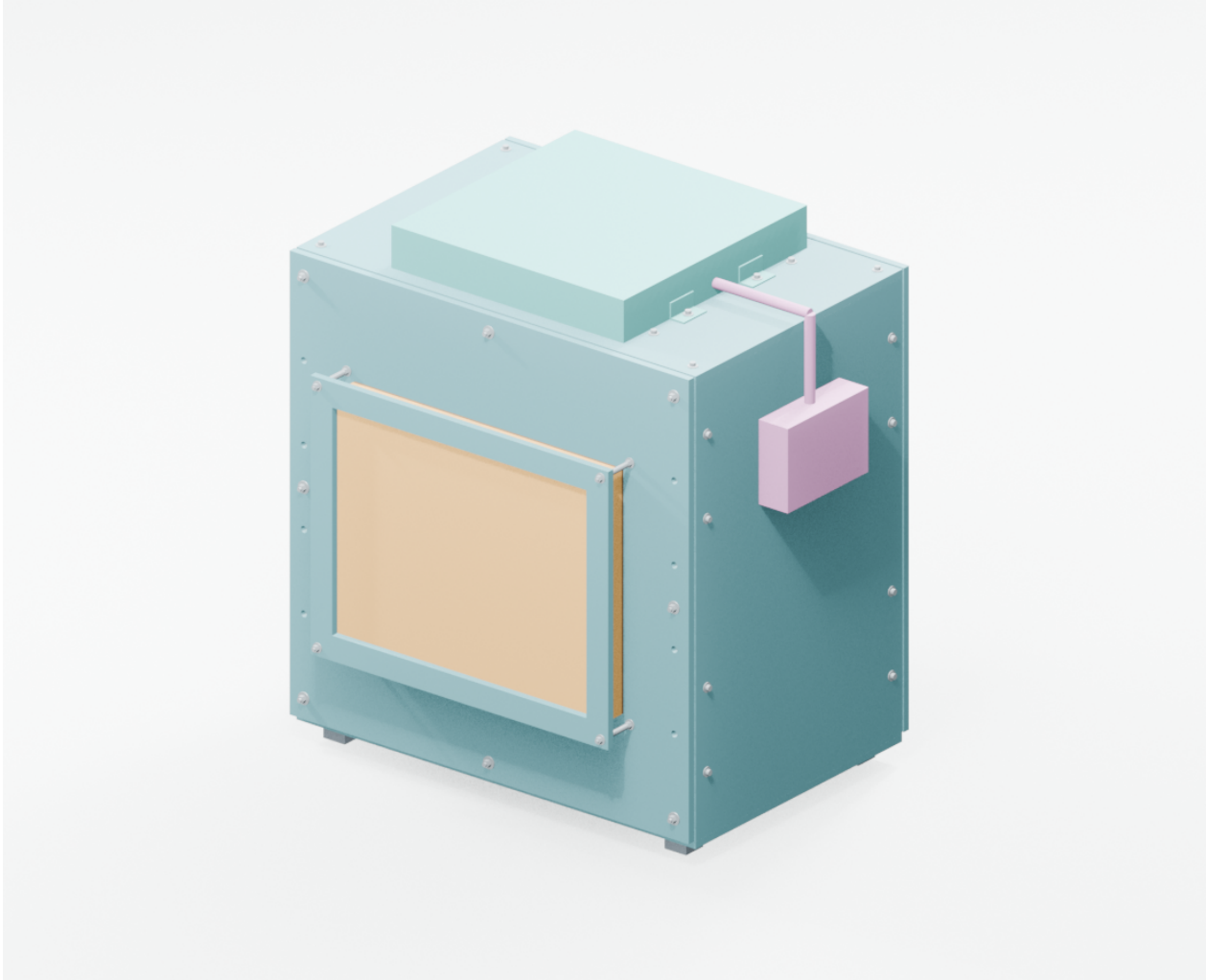


AQI Tower

Prototype development and funding review

We are preparing a small air-cleaning prototype. We need support to finish the engineering review, build it safely and measure whether it works.



Actual CAD-derived rendering, not a physical product photo. Purple parts reserve control-box and cable space. Guard openings are not represented accurately.

Current position: digital design exists; construction and performance are not yet approved. AQI Tower is a working name.

Report 1 Executive Summary

A short update for the college, NGO and funding team.

Vision for 2026

Build and safely test one indoor prototype. Use those results to decide whether a larger outdoor tower is worth developing.

Work completed

An integrated CAD model, mechanical drawings, an electrical reference plan, calculations and physical-test instructions are ready for review. No physical unit has been built.

Key decisions

Start with particle filtration indoors. Use a custom housing with two parallel filters and four fans. Keep water, gas-filter media and solar outside this first build.

Unresolved challenges

Actual filter resistance, sealing, guard strength, cabinet stability and electrical protection still need closure. Outdoor coverage is unknown.

Support requirements

Fund exact parts after approval, qualified design review, local fabrication, safe assembly and rented or paid measurement services. We do not assume a college lab.

Plan for upcoming month

Assign reviewers, resolve supplier inputs, agree test targets and collect quotes. Release and build only after approval. Choose a final project name. Dates depend on responses and workshop capacity.

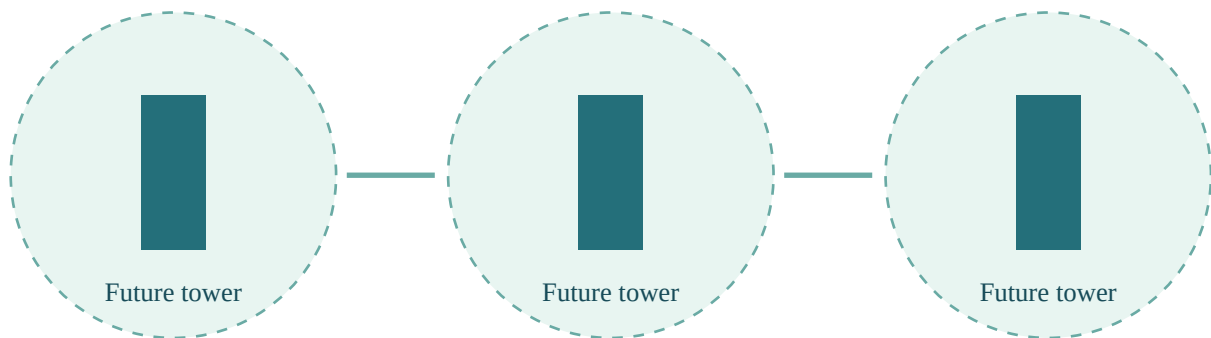
Report 2 Comprehensive Report

Vision for 2026

Build and safely test one indoor prototype. Use those results to decide whether a larger outdoor tower is worth developing.

Project idea and the clean air bubble

The long-term idea is a larger outdoor cylindrical tower. Each tower would aim to improve air near people. Several towers might form a chain of local cleaner-air zones.



Concept only. No measured radius, spacing or guaranteed protected zone.

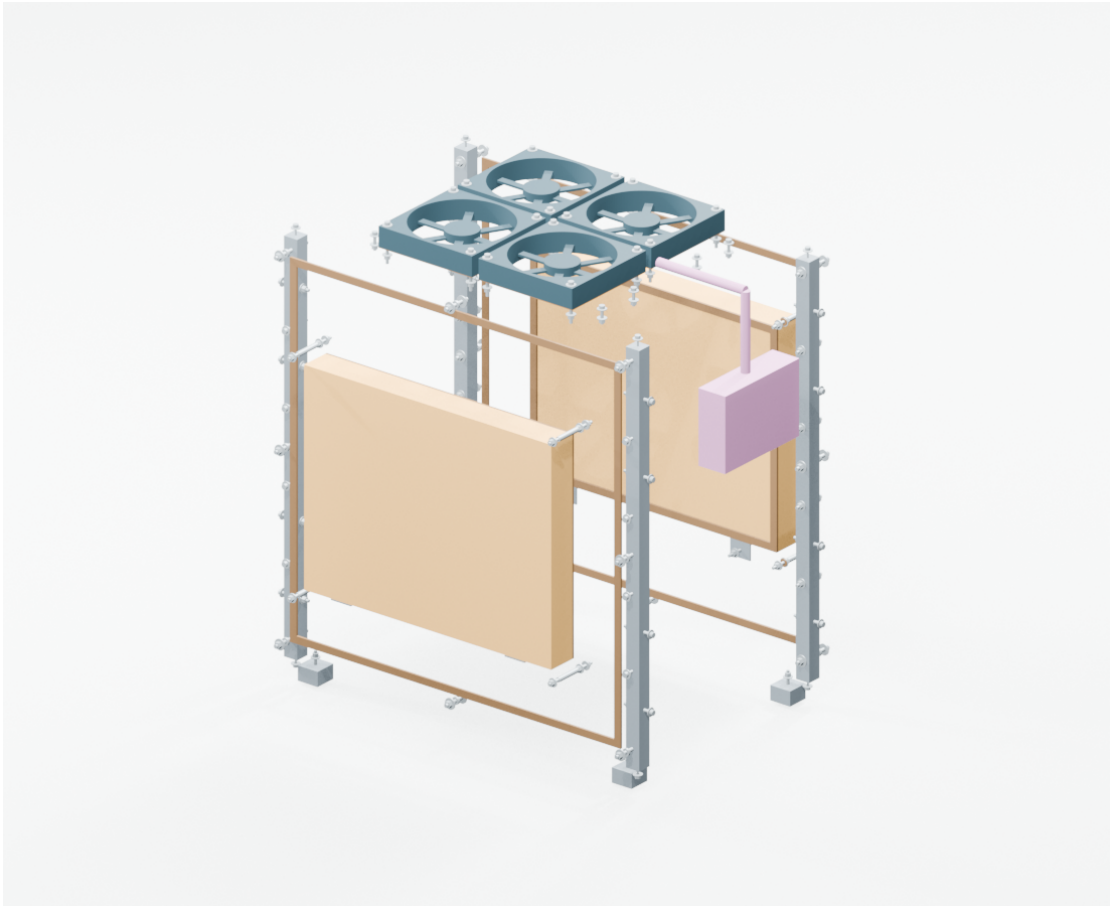
A bubble is a simple way to explain the goal. It is not a sealed dome or a proven clean-air boundary. Outdoor wind mixes air continuously, so we cannot promise a bubble size or cleaning percentage.

The current prototype is a small rectangular indoor unit, not the final outdoor tower. Keeping the first build simple helps us inspect, transport, measure and repair it.

Where it may be used later

Potential sites include sheltered waiting areas and other managed public locations. No site has been selected. Fully outdoor use needs separate heat, rain, dust, corrosion, anchoring and vandal-resistance design.

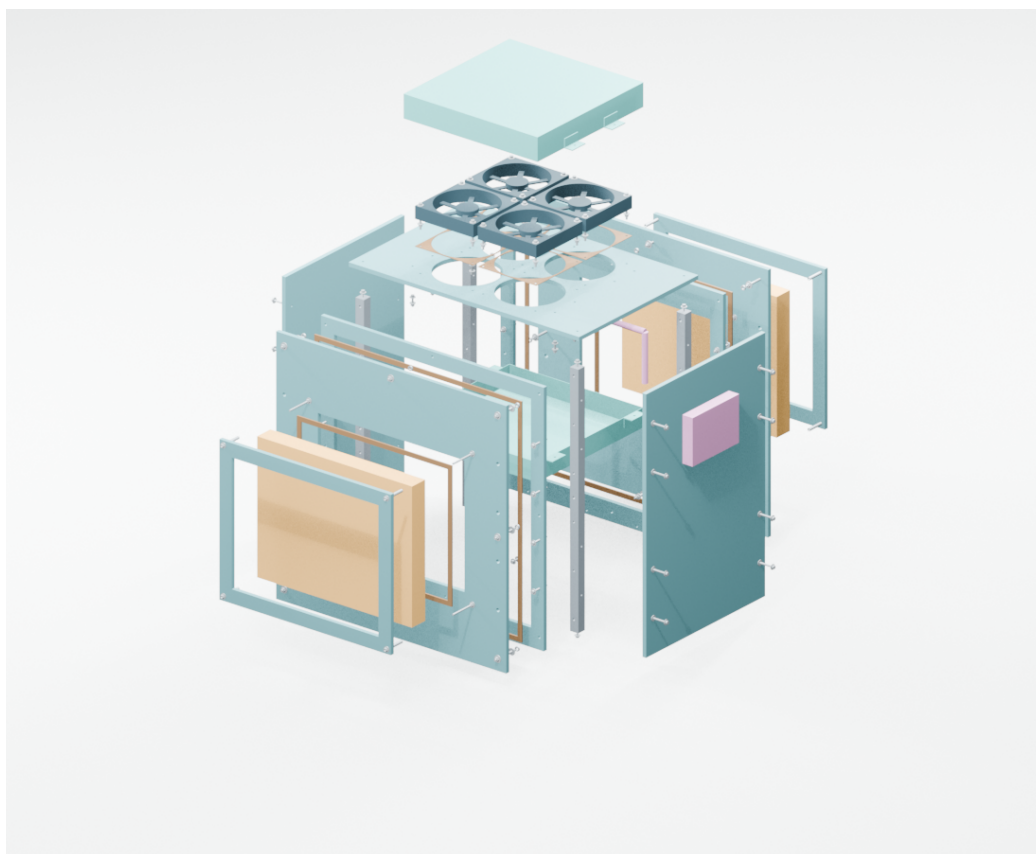
How the prototype works



Part	Simple explanation
Two filters	Air enters through two opposite filter branches. They work side by side, not in sequence.
Four fans	Pull air into a shared chamber and discharge it through the top.
Cabinet seals and guards	Keep the parts supported, limit leaks and restrict access to moving fans. These details still need approval.

Cutaway view removes panels and guards for explanation. Never operate an unguarded unit. Colours identify parts; they do not show measured pollution, flow speed or efficiency.

Work completed



The current model coordinates the cabinet, filters, seals, fan plate, guards and proposed hardware. Drawings, calculations, review sheets and test software are available for the internal team.

What the computer work tells us

We can check nominal fit and explore pressure losses. We cannot yet state actual airflow, cleaning efficiency, noise, power or filter lifetime. The fan and filter must work together under load.

Important evidence boundary

Earlier coloured CFD studies belong to a different large-tower geometry. They were partial or nonconverged and do not validate this prototype. The new animation explains assembly and the intended air path; it is not a new CFD result.

Key decisions and unresolved challenges

Key decisions

Start with particle filtration indoors. Use a custom housing with two parallel filters and four fans. Keep water, gas-filter media and solar outside this first build.

Question	Current answer
Has a tower been physically tested?	No. Physical prototypes built: 0.
Are parts and wiring released?	No. The integrated design is for review only.
Will it remove gases?	Not demonstrated. The first scope is particle filtration.
Does it have a verified HEPA rating?	No whole-device HEPA classification.
How large is the outdoor bubble?	Unknown. Outdoor benefit and spacing need field measurements.
Are costs and dates confirmed?	No. Quotes, supply times and fabrication capacity are still unknown.

The main missing work is component evidence, mechanical and electrical approval, then construction and controlled measurements. Another attractive model does not close those items.

Support requirements and upcoming month

Support requirements

Fund exact parts after approval, qualified design review, local fabrication, safe assembly and rented or paid measurement services. We do not assume a college lab.

Funding stage	What funding should enable
Review first	Resolve guard, structure, filter seal and electrical protection with named reviewers.
Build after release	Purchase approved parts and outsource the required cabinet and guard fabrication.
Measure after commissioning	Rent suitable instruments or commission a test service; record airflow, pressure, particles, noise and power.

Plan for upcoming month

Assign reviewers, resolve supplier inputs, agree test targets and collect quotes. Release and build only after approval. Choose a final project name. Dates depend on responses and workshop capacity.

The next measurable milestone

A review-approved prototype assembly followed by a safe first run with recorded airflow and pressure. A later controlled room test should compare background particle decay with purifier-on trials.

Supplier responses, fabrication, delivery, qualified commissioning and physical testing are separate tasks. No completion date or performance percentage is promised.

Use STAKEHOLDER_FUNDING.xlsx for staged funding discussions. Meetings content is omitted as requested; no meeting count has been invented.

Room size, time and what we target

The prototype is about 0.631 m high in the current display envelope. The viewer places a simple 1.70 m person beside it for scale. This is not an average Indian height or a picture of a tested installation.

The first target: airborne particles

We aim to reduce airborne dust and fine smoke particles, including PM2.5 and PM10. The actual capture and room reduction need measurements. Smoke gases, CO2, carbon monoxide, odours/VOCs and bacteria/virus control are not established. Particle filtration does not replace ventilation or removing pollution sources.

Why a bigger room takes longer

A room contains a volume of air: floor area multiplied by ceiling height. More air takes longer to clean at the same particle-cleaned air rate. The interactive viewer lets you change the area, height, assumed clean-air rate and elapsed time.

Example only	Ideal time to 90% fewer particles
25 m2 room, 2.8 m ceiling, assumed150 m3/h clean air	64.5 minutes
50 m2 room, same ceiling and assumed clean-air rate	128.9 minutes

These examples are NOT prototype ratings or safe-occupancy times. They assume perfect mixing, no new particles and no outdoor influx. Open windows, people, cooking, leaks and uneven mixing change the result.

A useful engineering requirement

For the example25 m2 room to reach90% reduction in30 minutes, the ideal model needs322.4 m3/h of particle-cleaned air. Even perfect capture at the assumed300 m3/h airflow screening point cannot meet that target: its ideal minimum is32.2 minutes. That airflow has not been measured. We must agree a realistic room/target before claiming success.

Sources checked5 October2026: EPA Guide to Air Cleaners in the Home (epa.gov/indoor-air-quality-iaq/guide-air-cleaners-home), and CDC Appendix B Air (cdc.gov/infection-control/hcp/environmental-control/appendix-b-air.html). The latter supplies the ideal decay equation/mixing limitations, not a medical approval for our prototype.